RNA-seq and analysis of differential gene expression

Slides adapted from: Tom Booker & Bernadette Johnson
Bioinformatics for Evolutionary Biology and Conservation

Outline

1. Background
2. Overview of the methods and workflow
3. Quantifying expression levels
4. Analyzing patterns of expression
5. Technical considerations

Learning outcomes

Explain how RNA-seq is generated and used

Identify the basic steps to align and analyze RNA-seq data

Introduction and background

Why use RNA-seq?

Can you think of some uses for RNA-seq?

Introduction and background

Why use RNA-seq?

```
>NC_090847.1:15049400-15049553 Syngnathus scovelli strain Florida chromosome 1, whole genome shotgun sequence
AAACAAGGAATTTGACTTCGGTAAATCACAGCCTCTGTTCAACATTTAGGTGACTAACAACAACACTCAGGACATGTGAA
GAACGAAAGATATTCTCAAACACCCCCTGATCTTAAACTCCCAAGAGGGCAAGGAAAAACTCAAAACTCCAGCT
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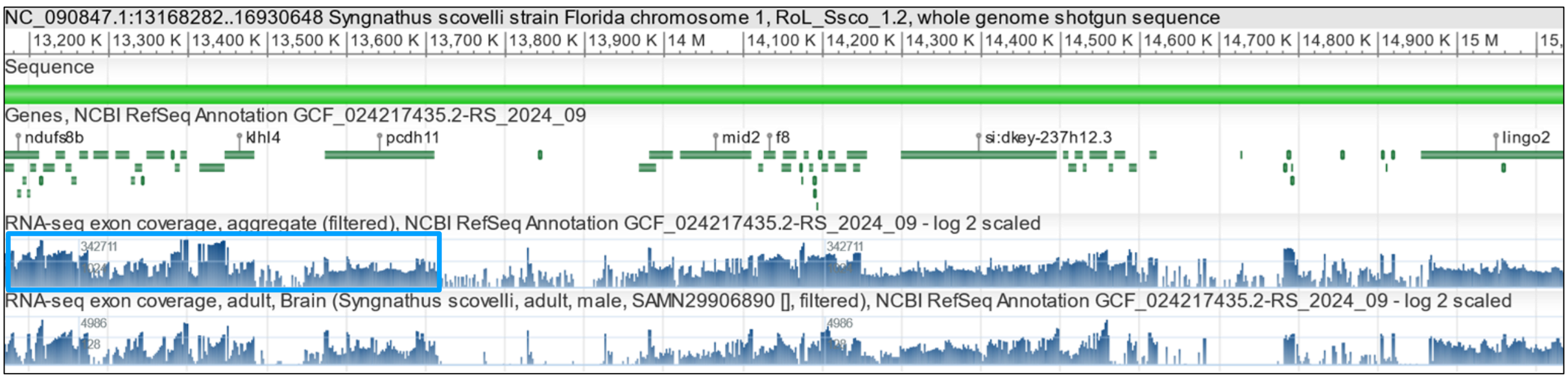

Genome annotation: Identifying transcribed regions of the genome

Introduction and background

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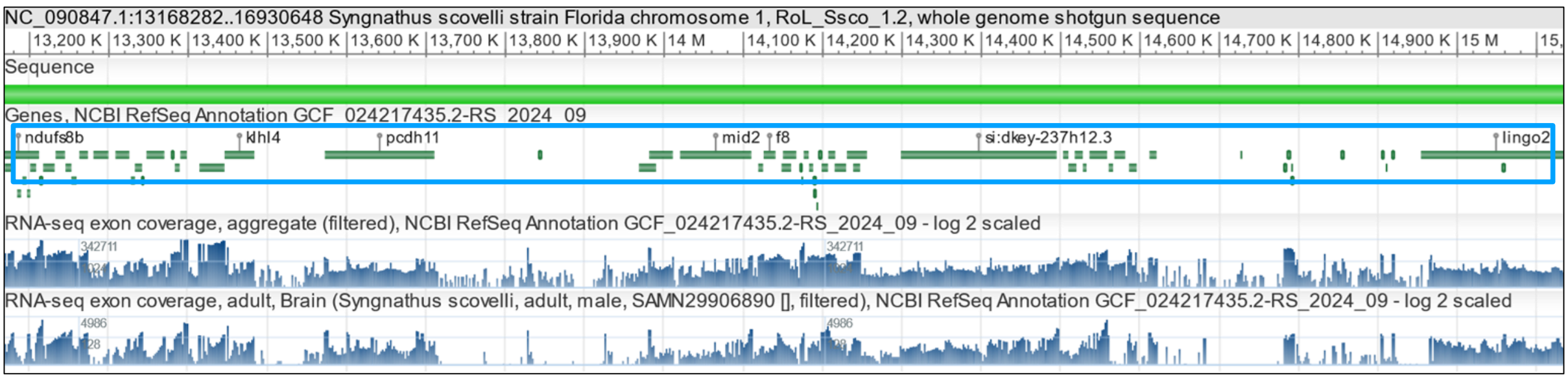
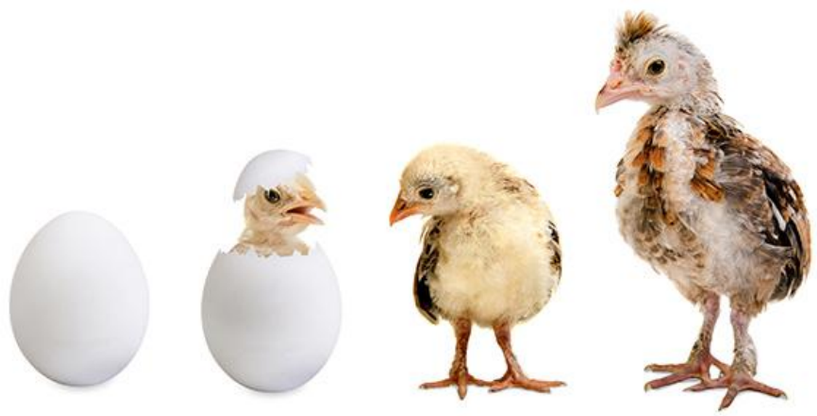


Genome annotation: Identifying transcribed regions of the genome

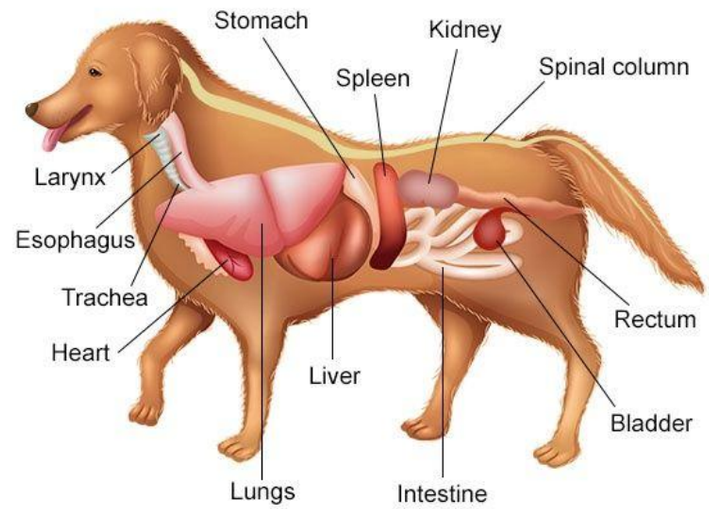
and designating locations of exon-intron boundaries (splice junctions)

Introduction and background

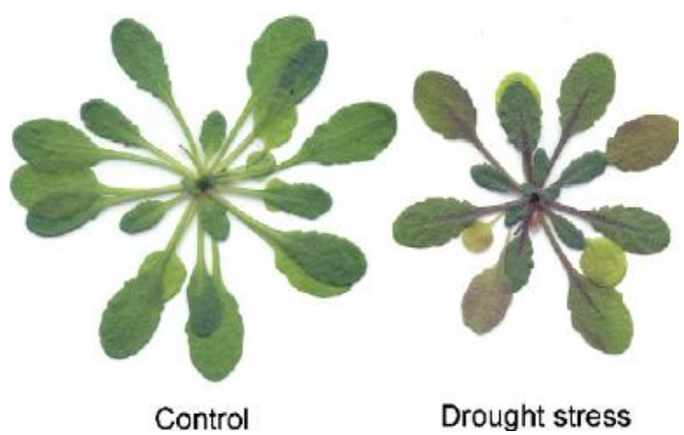
Quantifying differences in gene expression



Developmental timepoints



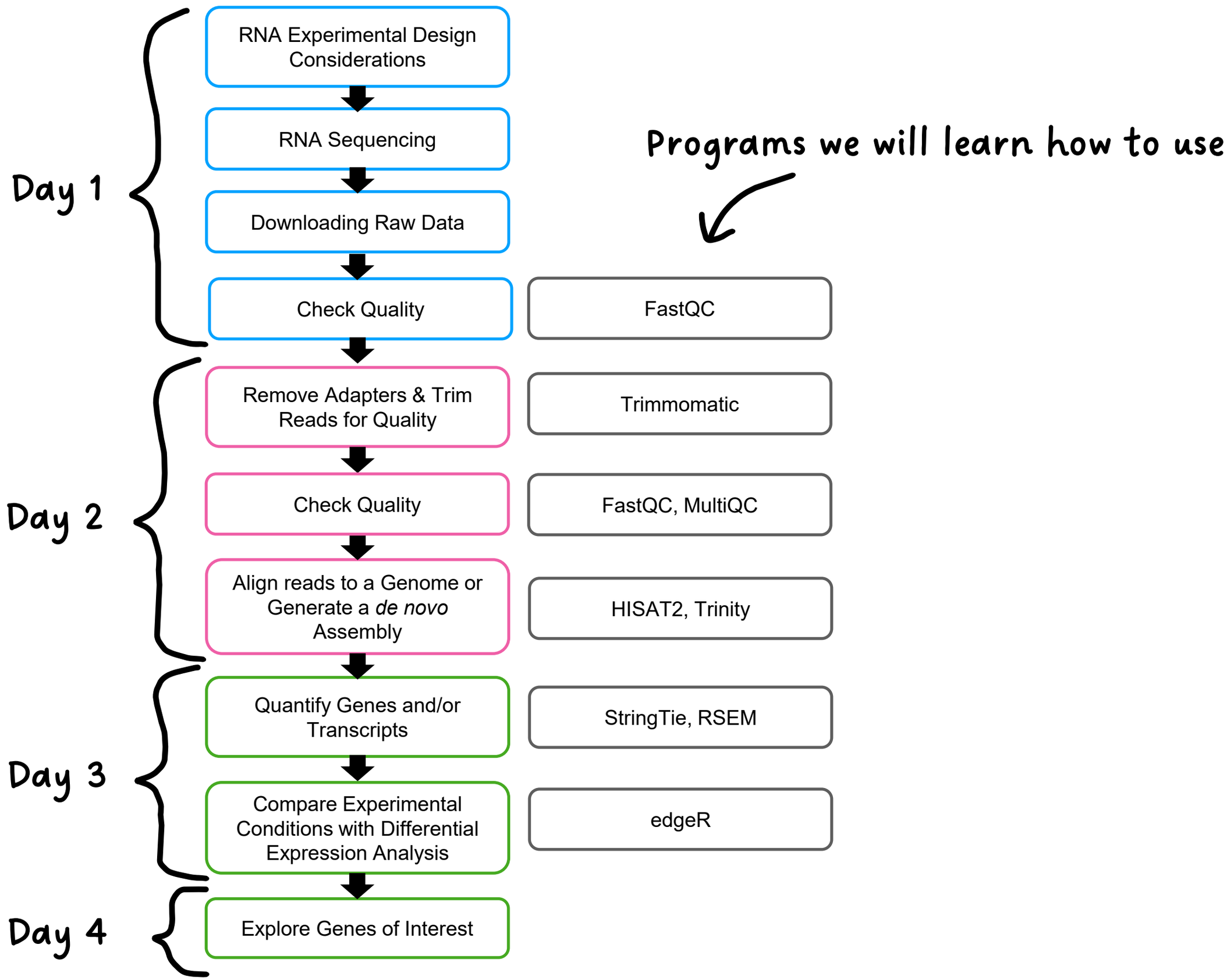
Different organs, tissues, or cell types



Experimental treatments

Between sexes, ecotypes, morphs

Basic RNA-seq pipeline



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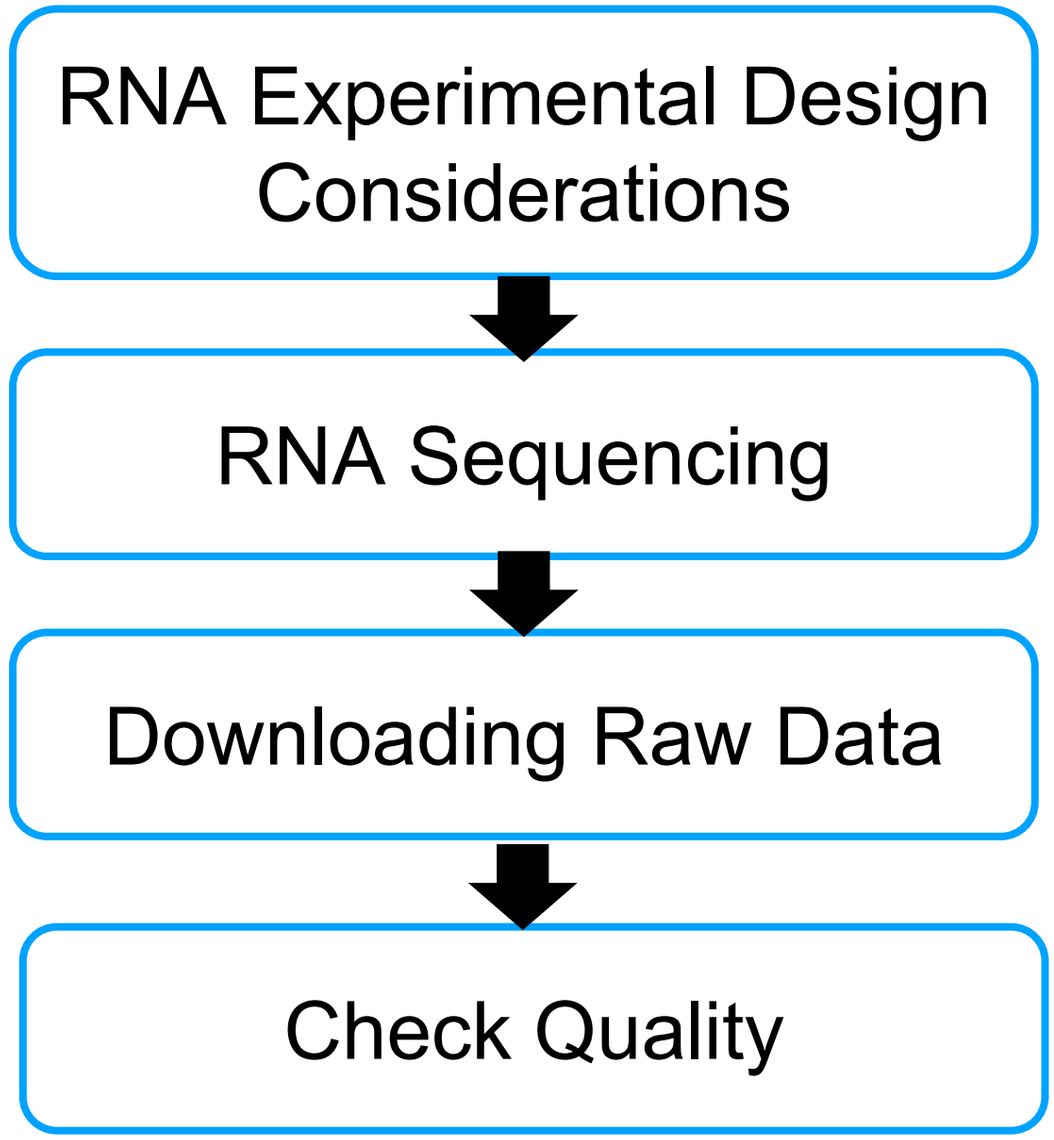
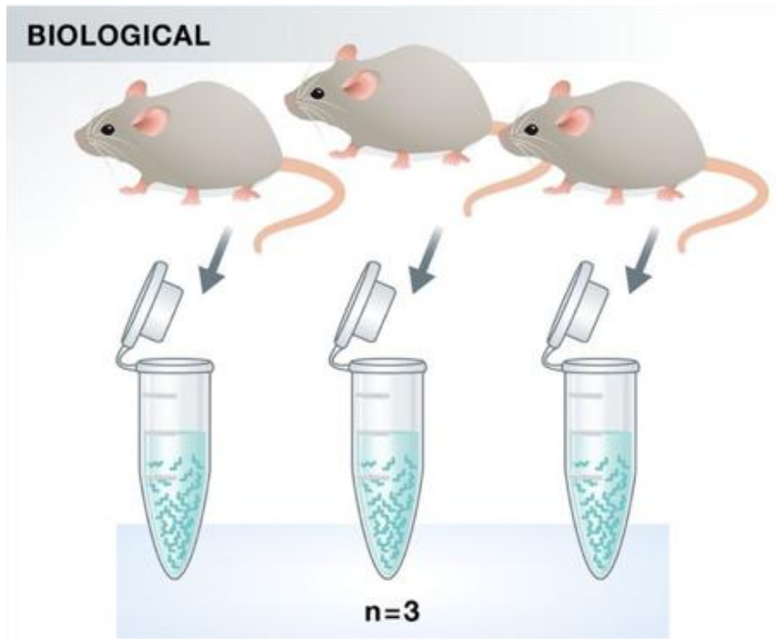


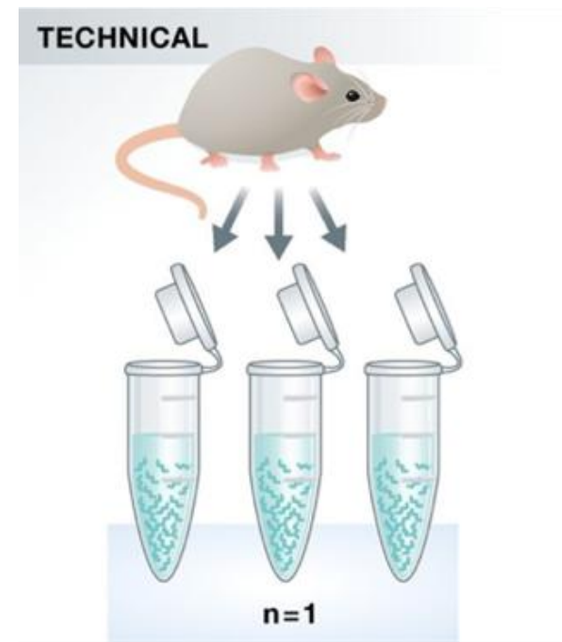
RNA Experimental Design Considerations

- Types of Sampling Strategies
- Sequencing Depth
- Batch Effects
- Other Types of RNA Sequencing Methods

Types of Sampling Strategies

Biological vs Technical Variation





- Multiple samples capture biological variation.
- Usually more important than technical sampling.

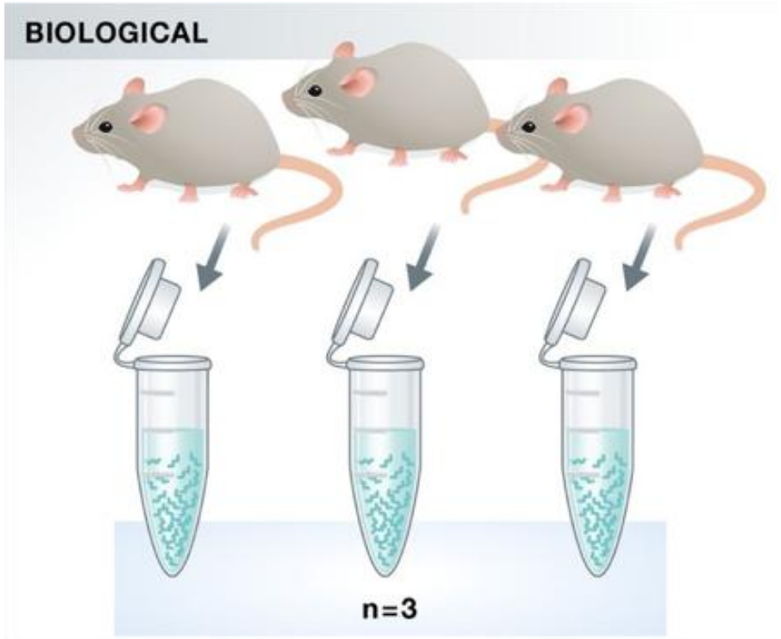
- Multiple of the same sample.
- Increased sequencing depth.

https://hbctraining.github.io/Intro-to-RNA-seq-hpc-salmon/lessons/experimental_planning_considerations.html

Types of Sampling Strategies

How to minimize biological heterogeneity?

- Subsampling tissue (micro dissections)
- Sampling only one cell type (scRNA)
- Accounting for potential sources of variance within samples (might be species-specific, like circadian rhythm, developmental age, morph)



https://hbctraining.github.io/Intro-to-RNA-seq-hpc-salmon/lessons/experimental_planning_considerations.html

Sequencing Depth

What is sequencing depth?

- It is the total reads or fragments per sample.
- Usually samples are sequenced for 20-50M reads.

Why do we care about sequencing depth?

- It reflects the portion of transcriptome captured (determines how much diversity is captured).
- Since RNA-seq is a relative measurement, you are looking at the proportion of reads counted per sampling unit. This means you need more sampling units to capture genes with low expression.

Sequencing Depth

How do I determine my ideal sequencing depth?

Depends on:

- transcriptome size (how many protein coding regions of genome)
- if you anticipate lots of rRNA contamination
- if you are interested in lowly expressed genes
- if you are interested in alternative splicing (isoform analysis)
- if you want to do an allele-specific expression

What about coverage?

- For RNA seq, depth is typically more useful than coverage (the proportion of the genome has been sequenced). A large portion of the genome may not be expressed.

Sequencing Depth

Sample Number vs Sequencing Depth

Increasing sample size, generally:

- allows for the identification of more unique reads
- provides more robust expression profiles than increasing depth
- gives more statistical power than increasing depth

Increasing sequencing depth beyond 20M reads make less of an impact (for differential expression analyses).

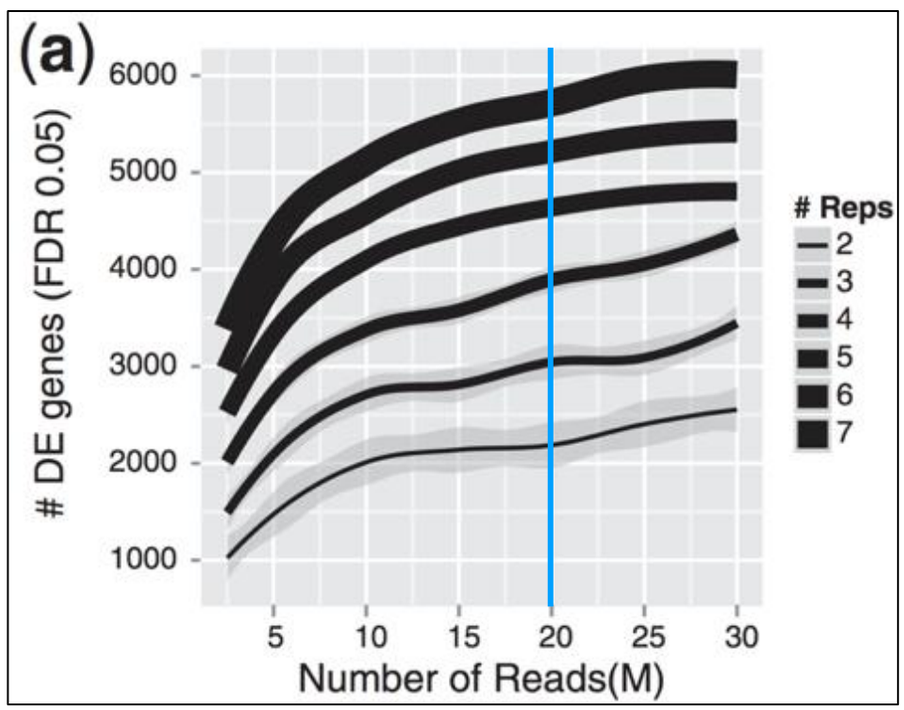


Table 1 Statistical power to detect differential expression varies with effect size, sequencing depth and number of replicates

	Replicates per group		
	3	5	10
Effect size (fold change)			
1.25	17 %	25 %	44 %
1.5	43 %	64 %	91 %
2	87 %	98 %	100 %
Sequencing depth (millions of reads)			
3	19 %	29 %	52 %
10	33 %	51 %	80 %
15	38 %	57 %	85 %

https://hbctraining.github.io/Intro-to-RNA-seq-hpc-salmon/lessons/experimental_planning_considerations.html

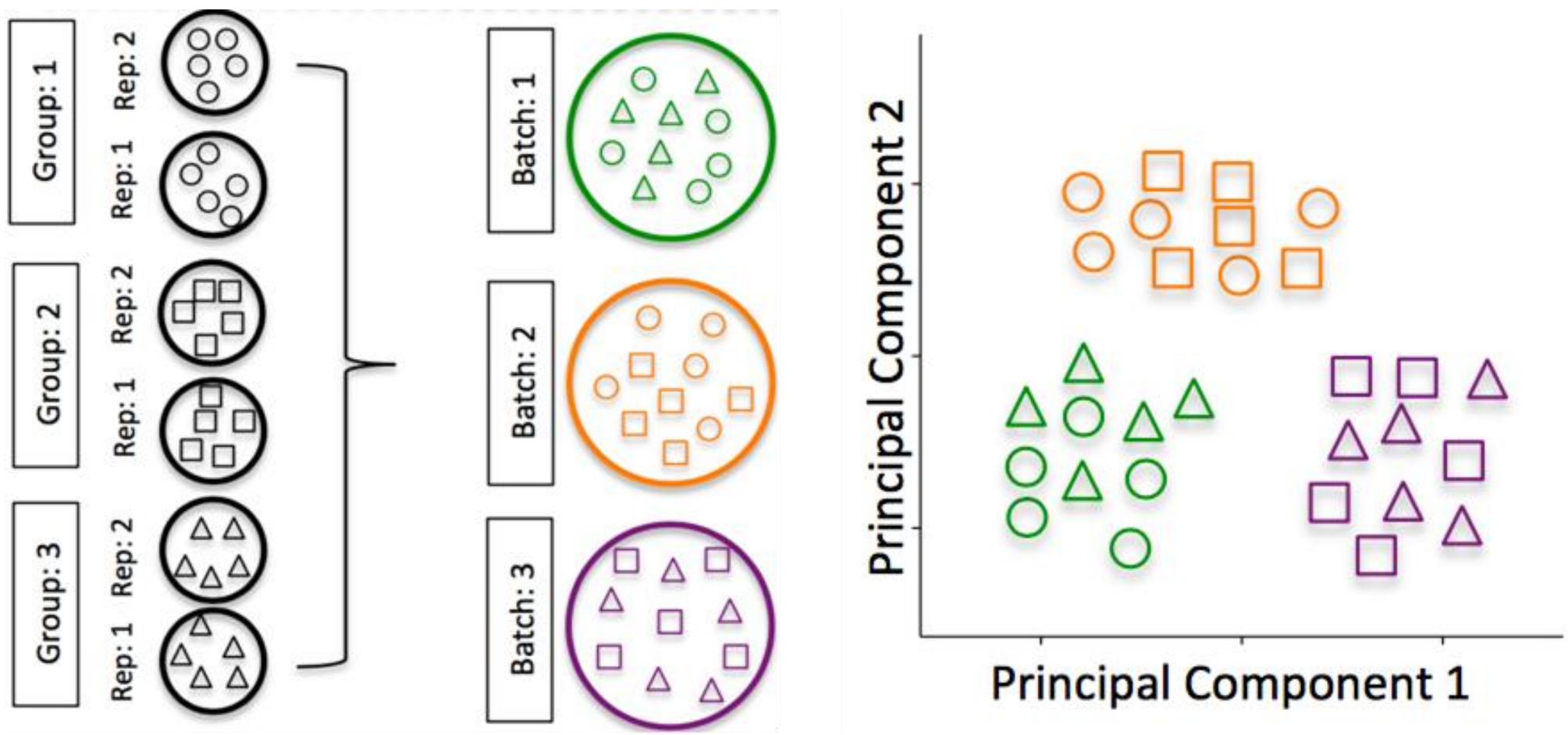
Batch Effects

What are batch effects?

- Non-biological technical artifacts that create variation in data.

What causes batch effects?

- Different lab conditions, difference in person performing the extractions, differences in PCR runs during library prep, enzymatic efficiency, can be anything!



Hicks SC, et al., bioRxiv (2015)

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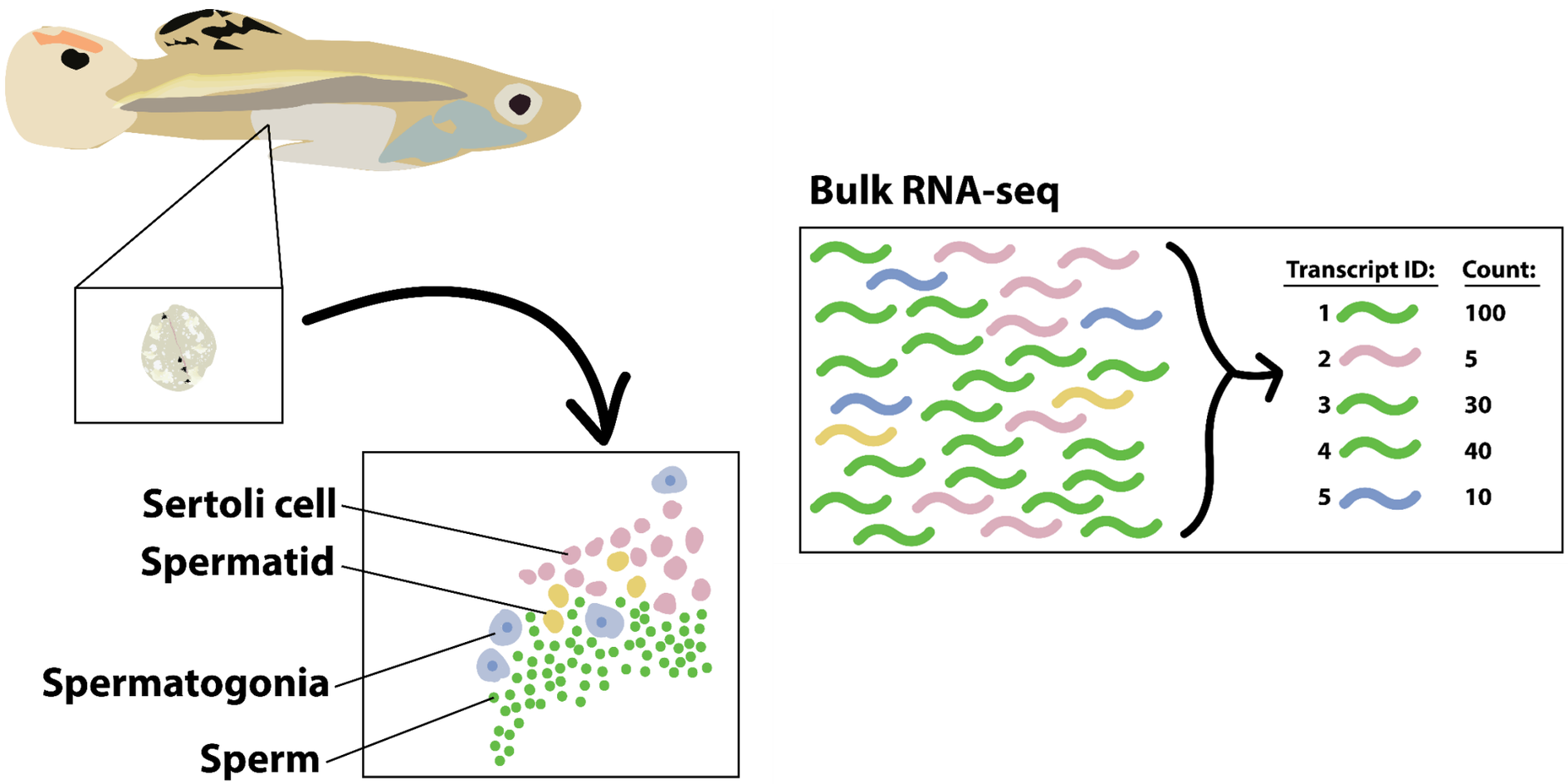
How can I mitigate batch effects?

- Randomize replicates during sample prep and sequencing (RNA extraction, library prep and sequencing).
- If you cannot randomize, there are some programs that can correct batch effects (can be done in DE programs like edgeR or DESeq2).

Other Types of RNA Sequencing Methods

Single cell sequencing: Label transcripts on a cellular level



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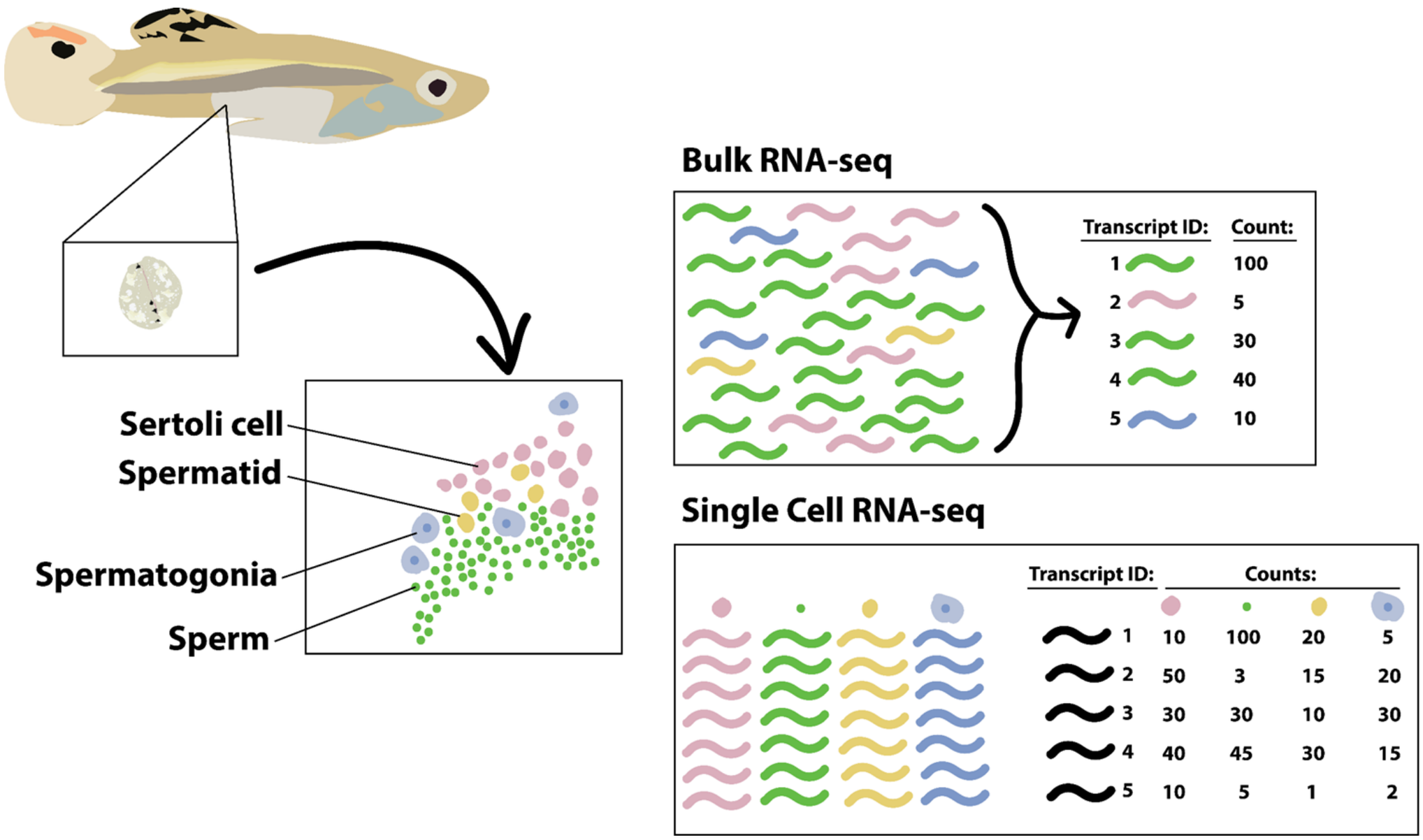
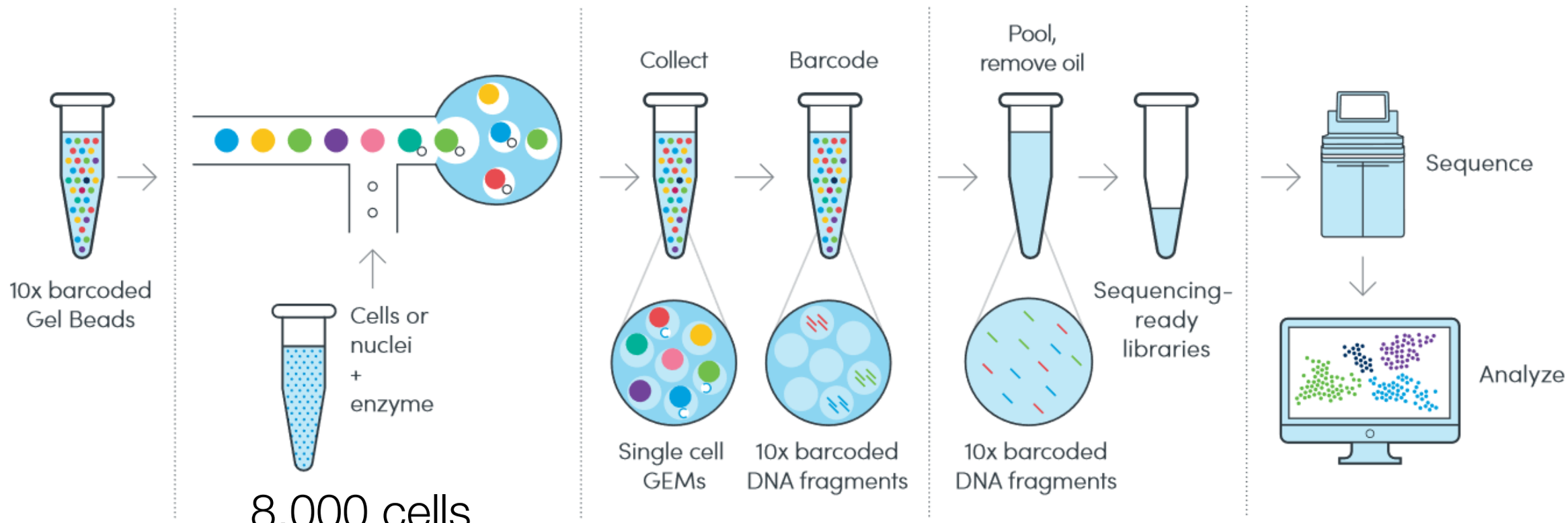
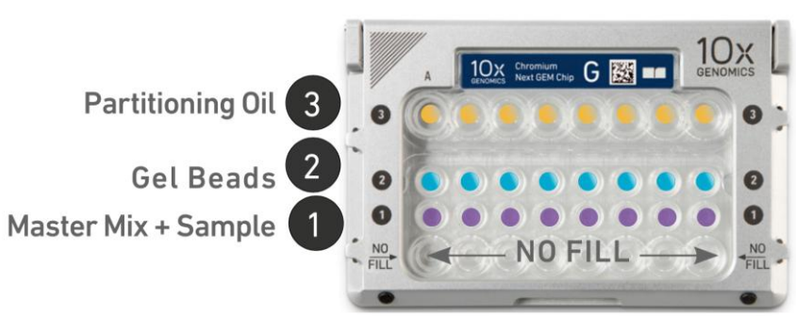


Benefit: provides cell-level transcriptome, normalizes for differences in cell abundances between samples (developmental vs expression differences).

Other Types of RNA Sequencing Methods

Single cell sequencing: how it works

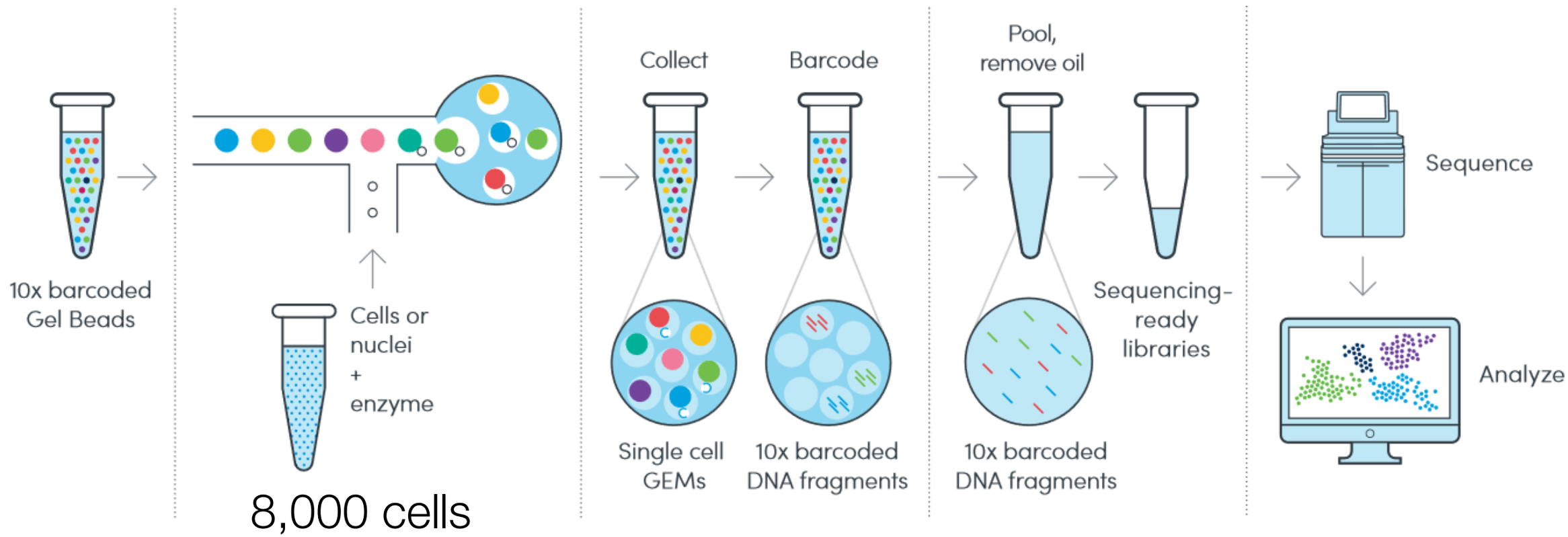


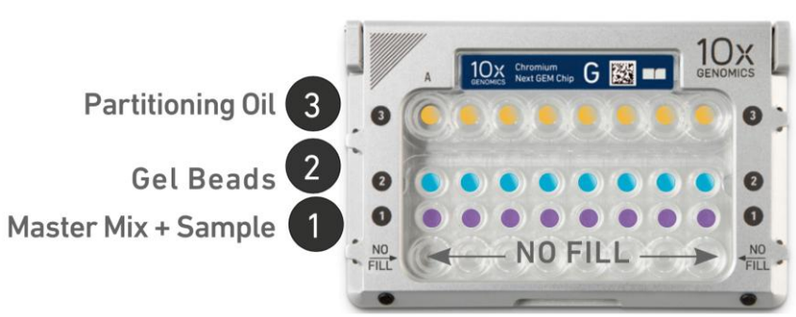


10xgenomics.com/blog/single-cell-rna-seq-an-introductory-overview-and-tools-for-getting-started

Other Types of RNA Sequencing Methods

Single cell sequencing: how it works





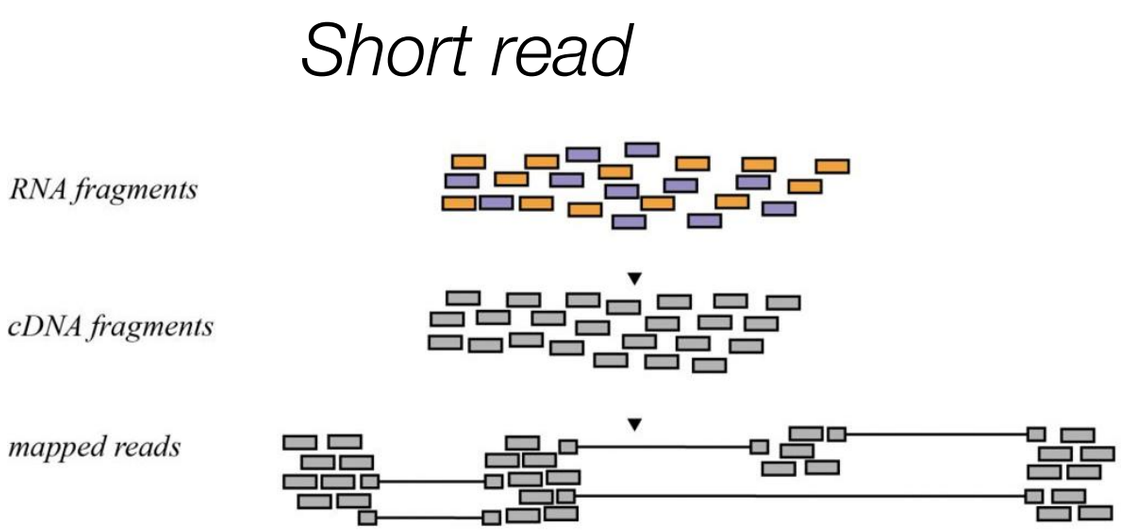
~1,600 CAD per sample
~25,000 CAD for instrument

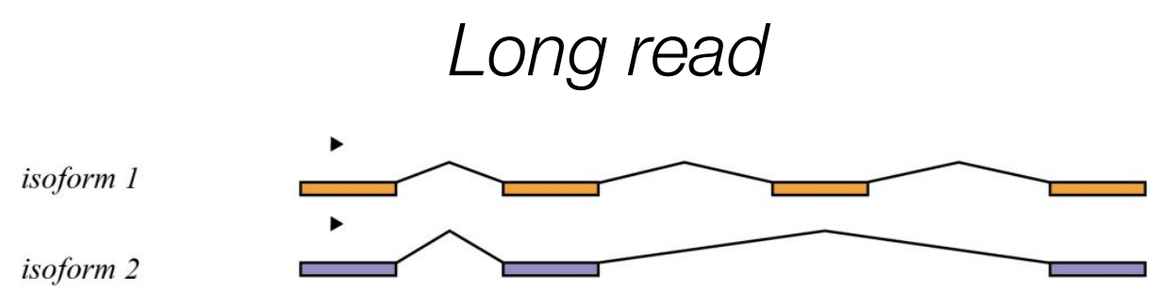
Cons: very expensive, requires specialized equipment, very sensitive to failure

10xgenomics.com/blog/single-cell-rna-seq-an-introductory-overview-and-tools-for-getting-started

Other Types of RNA Sequencing Methods

Long read transcriptome sequencing (e.g., PacBio) is an alternative (no assembly required, more confident isoform discovery).





RNA extractions are similar to bulk (somewhat easy) but can be more expensive (~4,000 CAD per sample for library prep and sequencing)

RNA Sequencing

- What happens during library preparation?
- What are adapters?
- What does demultiplexing mean?

Library Preparation

What happens during library prep?

- RNA is isolated, fragmented, and cDNA is synthesized
- Illumina specific adapters are added to the ends of your DNA molecule
- Sometimes ribosomal RNA (rRNA) is intentionally reduced with either polyA enrichment or rRNA depletion

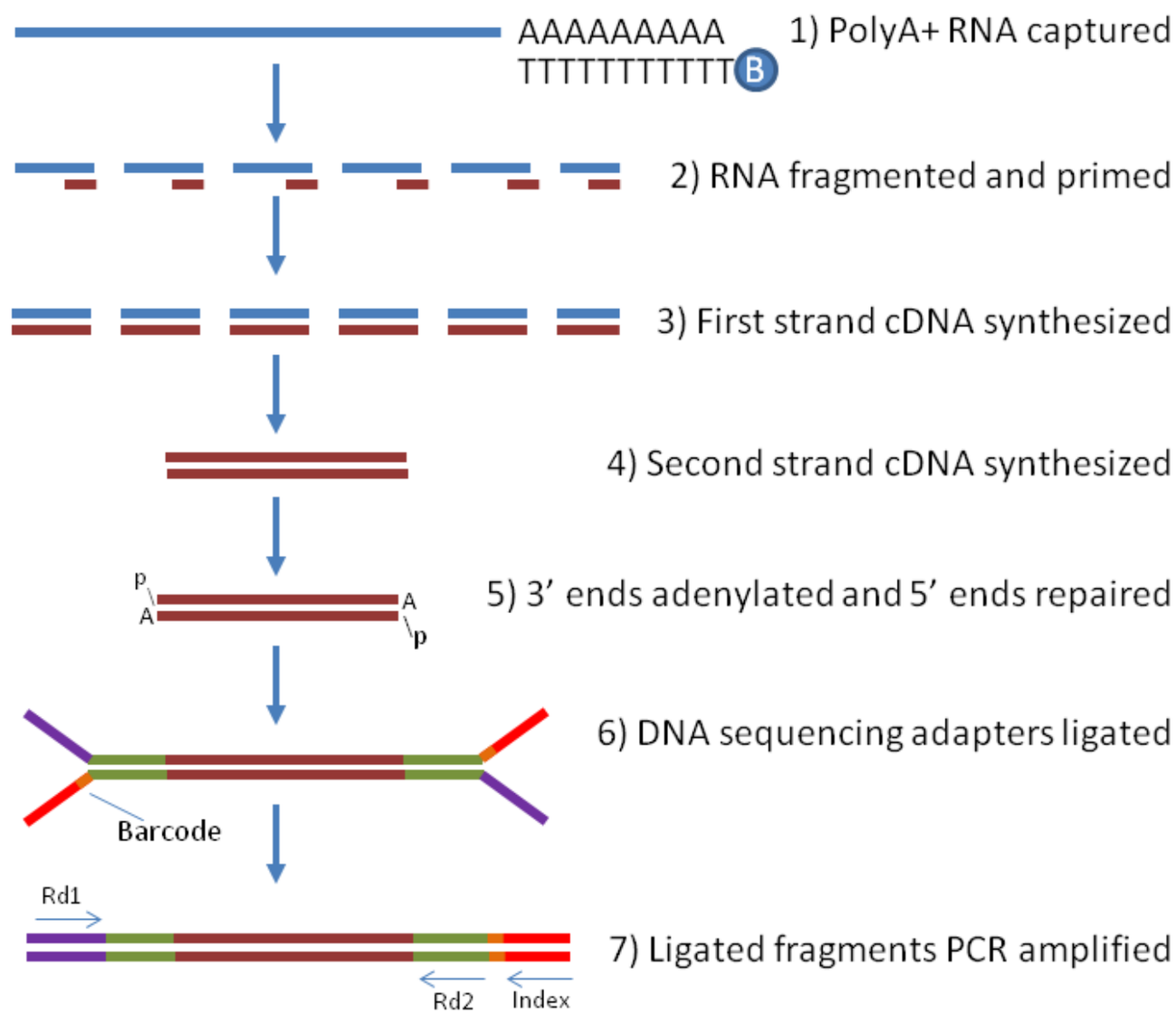
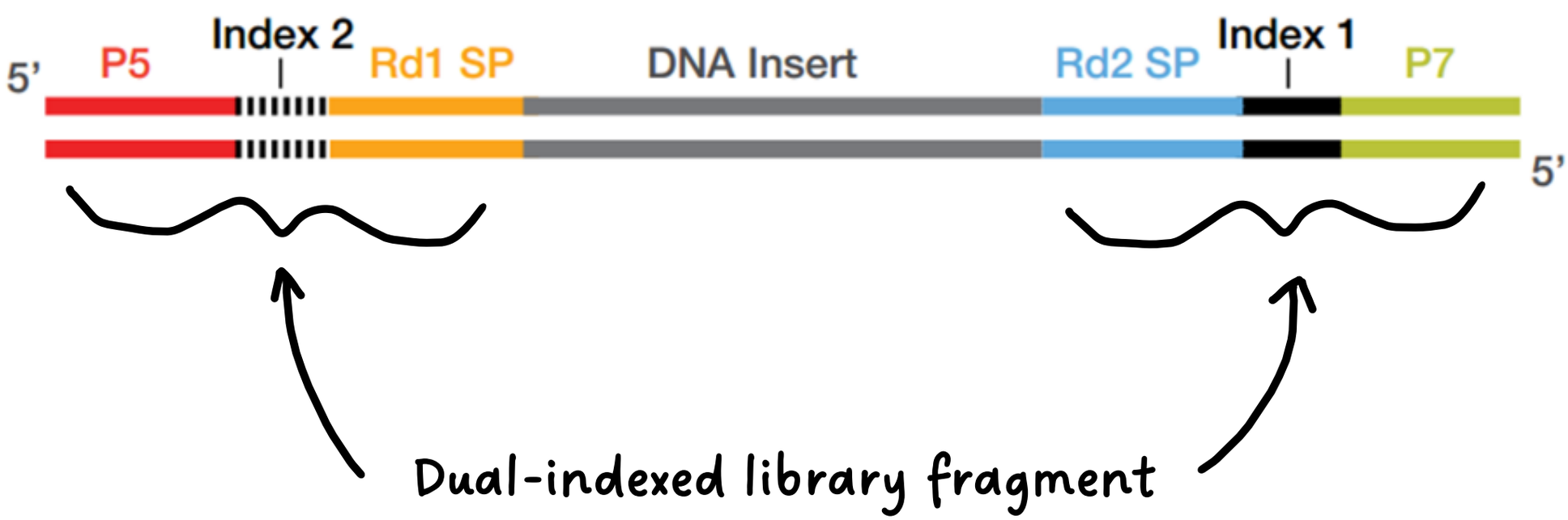


Figure from http://aws.labome.com/

Adapters

What are adapters?

- Adapters are short, synthetic oligonucleotide sequences ligated to the ends of DNA fragments during library prep



https://support.illumina.com/ko-kr/bulletins/2020/06/illumina-adapter-portfolio.html

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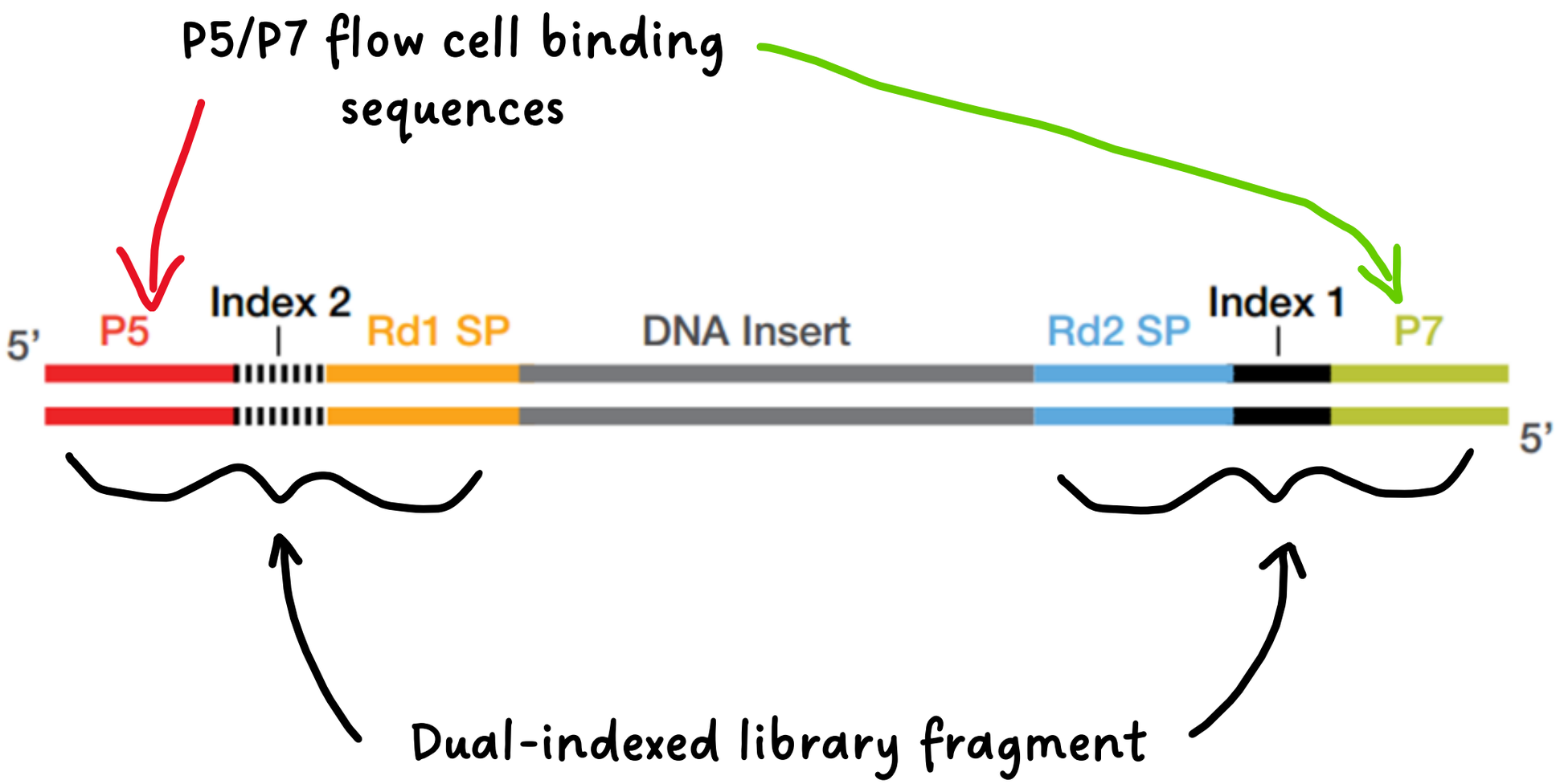
They contain many parts:



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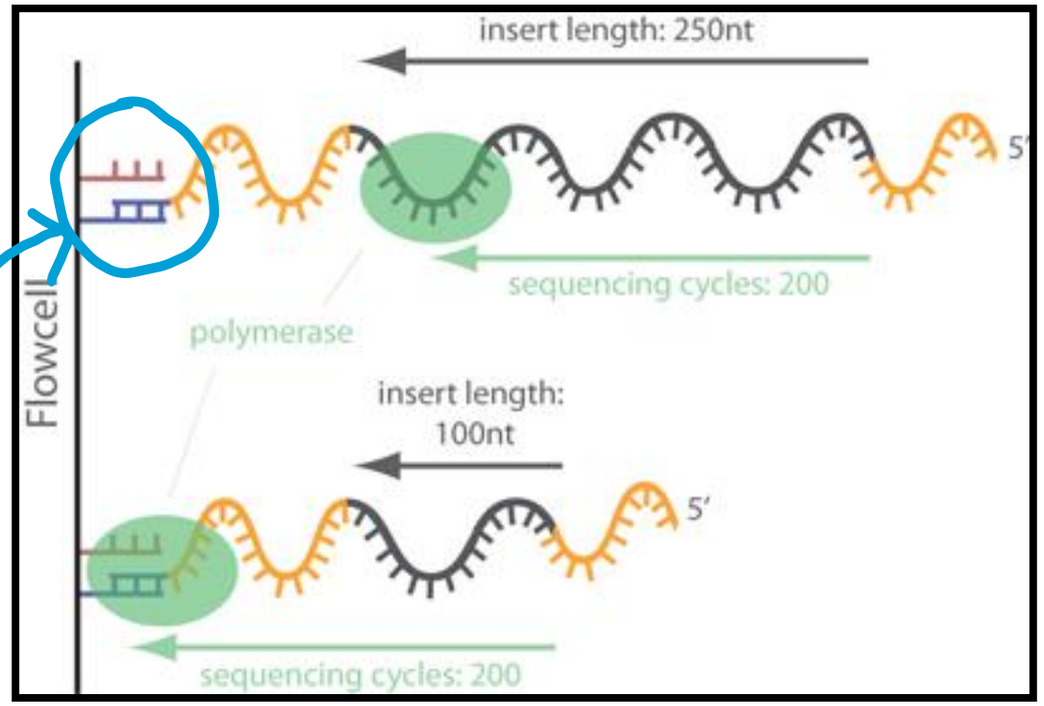
They contain many parts:

P5/P7 flow cell binding sequences =

provides a matching binding sequence for the DNA to bind onto the flowcell



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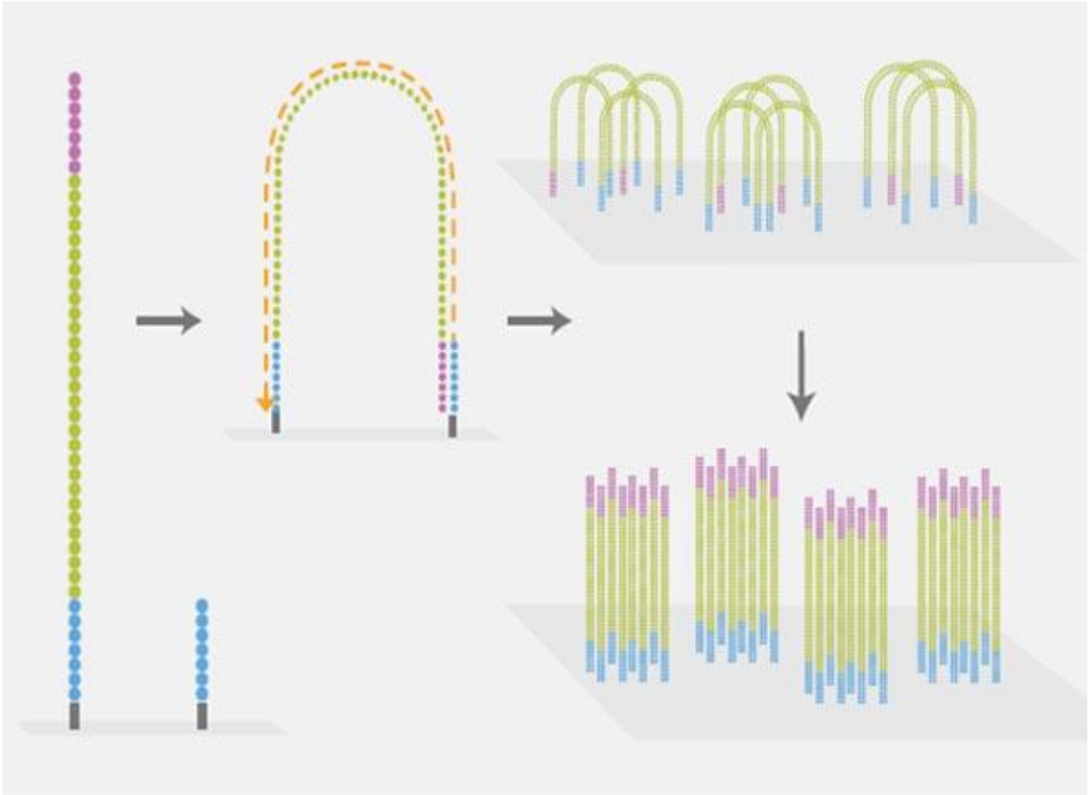

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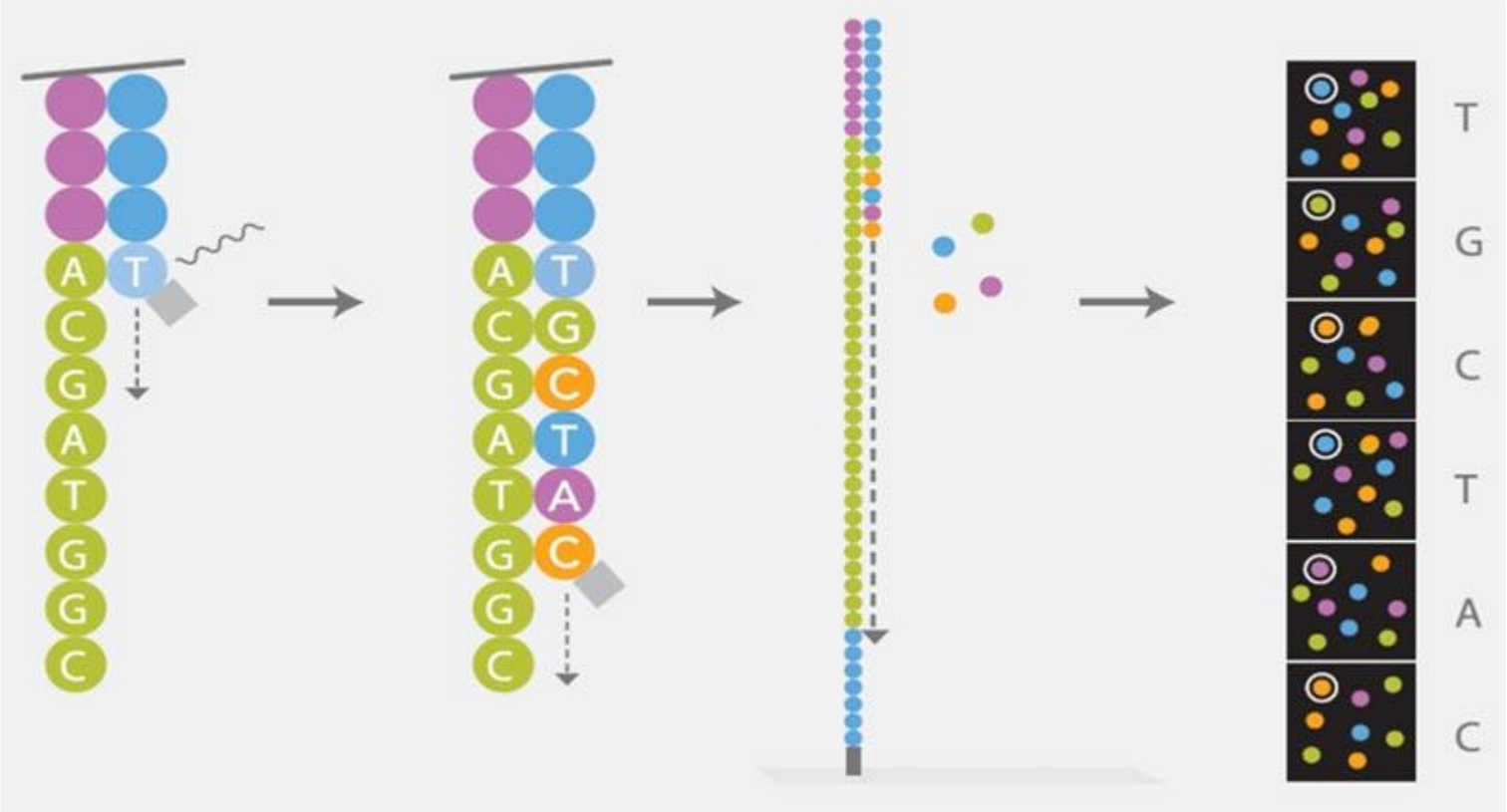
gives clear light color flash to identify between ATCG



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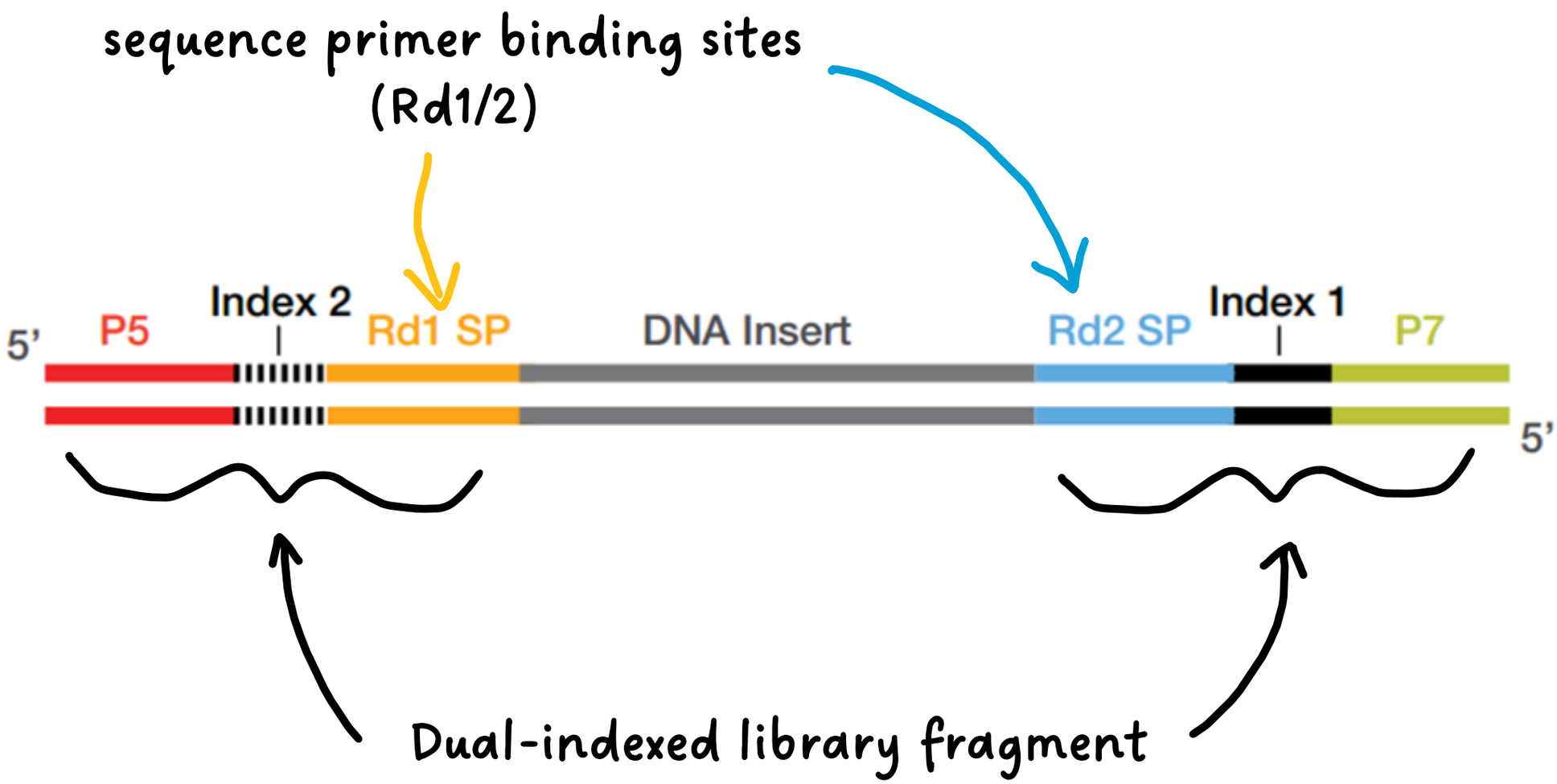

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Adapters

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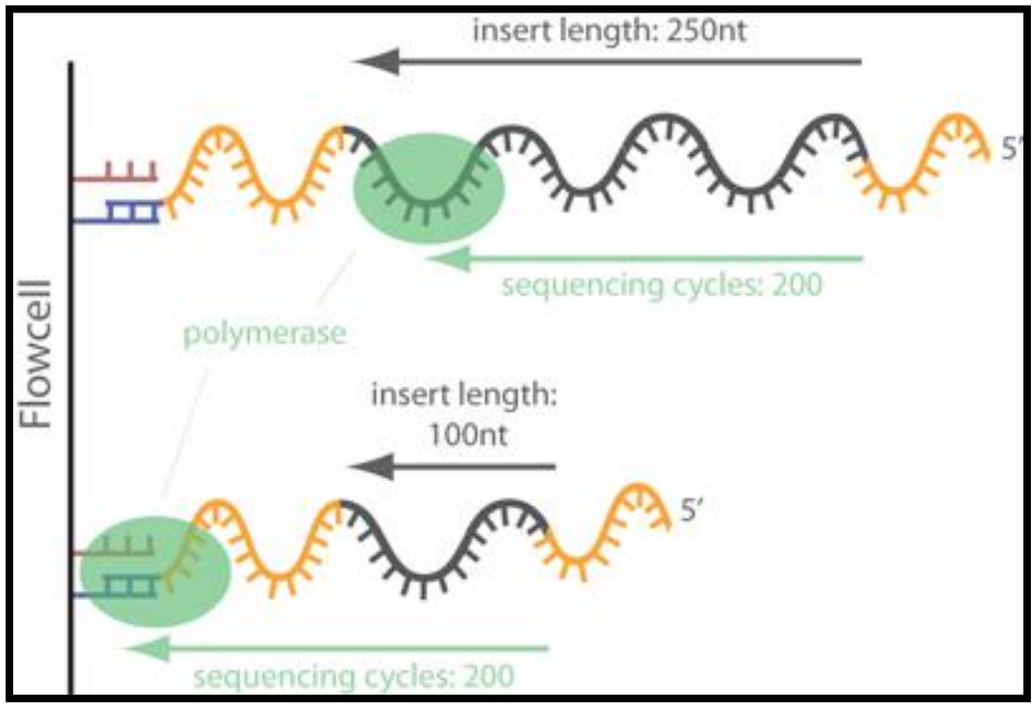
sequence primer binding sites (Rd1/2) =

initial sequencing by allowing sequencing primer and DNA polymerase to bind



https://www.ecseq.com/support/ngs/trimming-adapter-sequences-is-it-necessary

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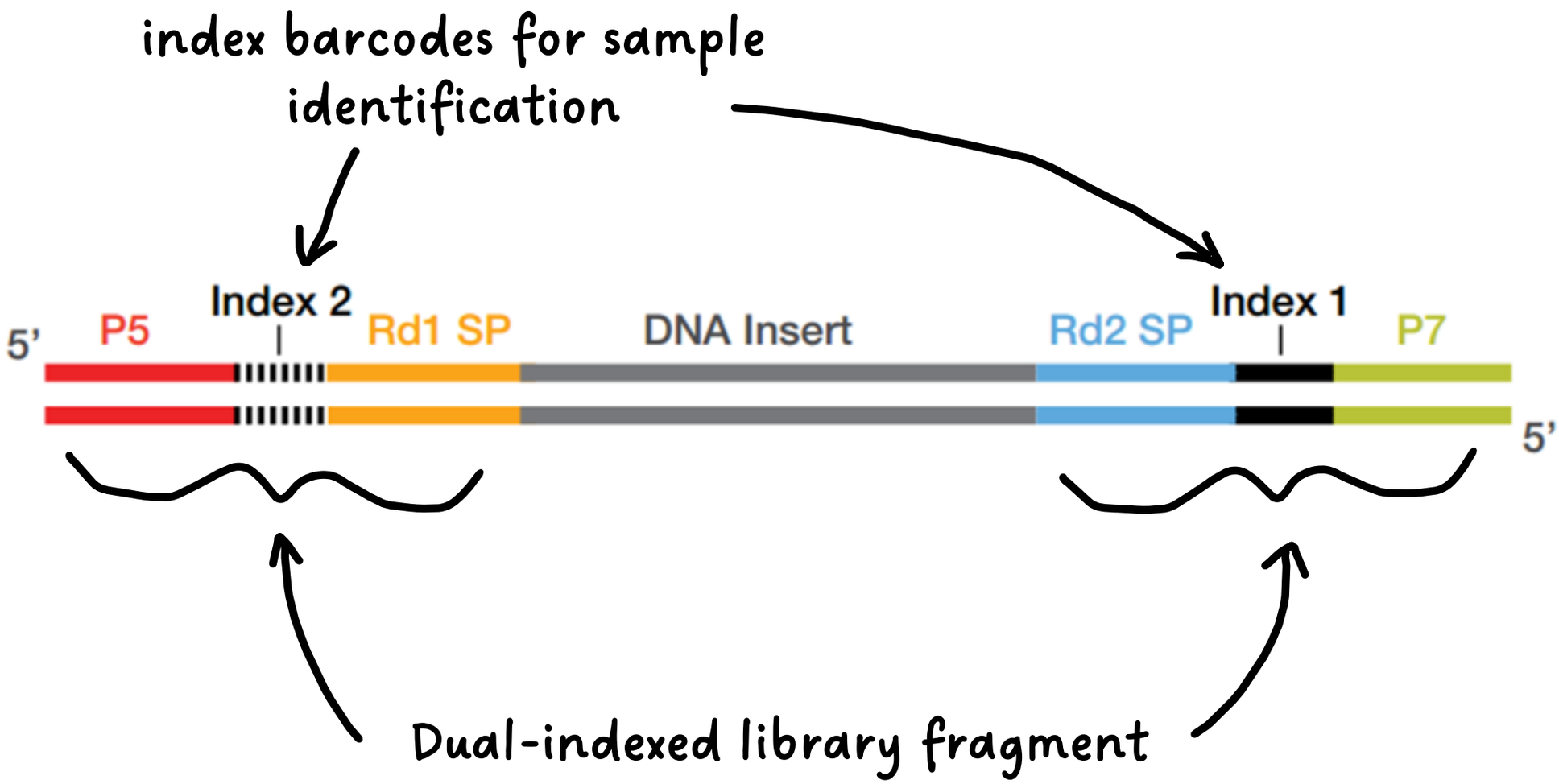

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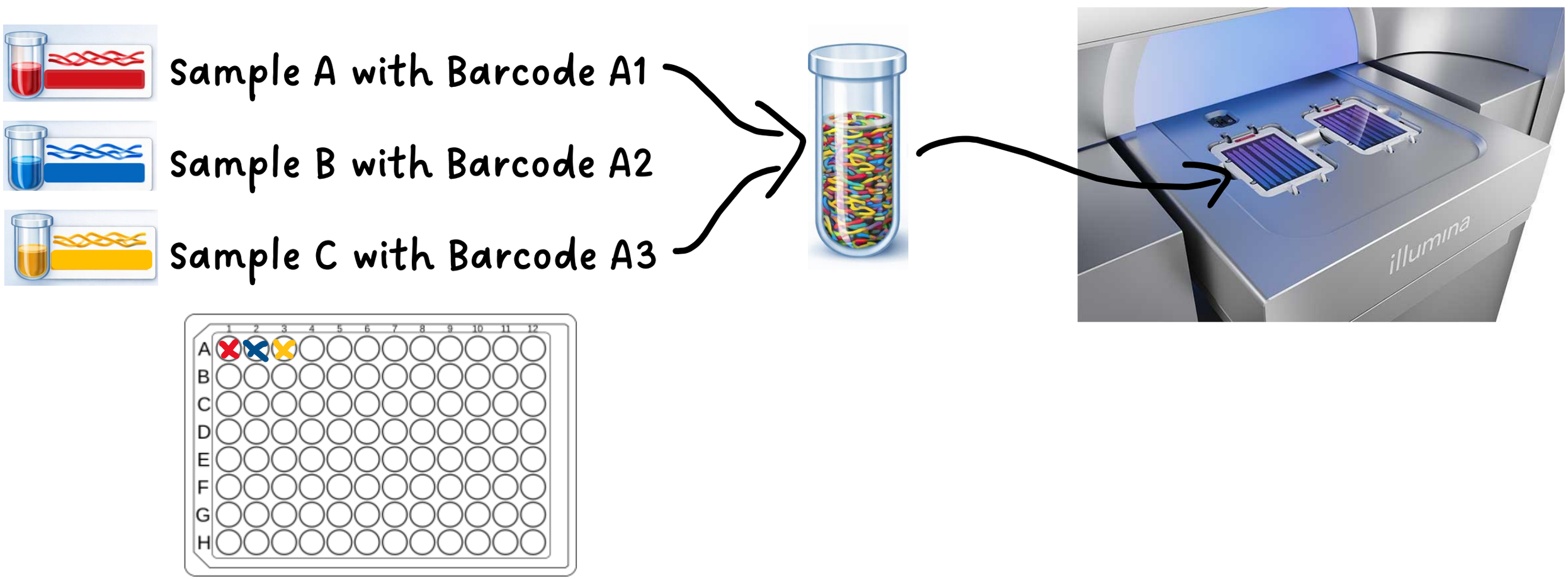
Demultiplexing

What is multiplexing?

- When barcodes or indices are added during library prep, and the samples are pooled together then loaded onto a chip to be sequenced.



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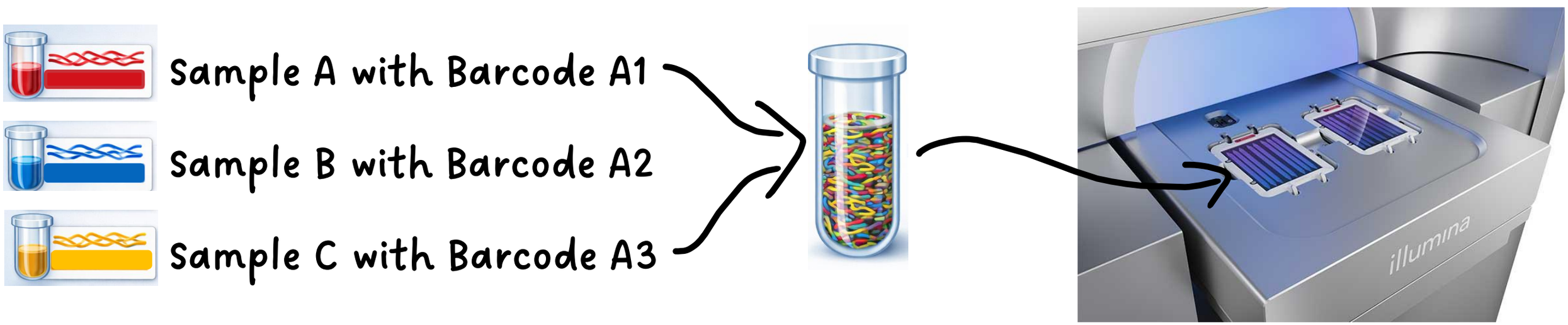


Why multiplex?

- Modern Illumina lanes generate far more reads than most single samples require, so multiplexing allows multiple indexed samples to share a lane and reduces cost per sample.

- Distributing samples across lanes prevents batch effects from lane-specific variation (in case one lane is more efficient than another).

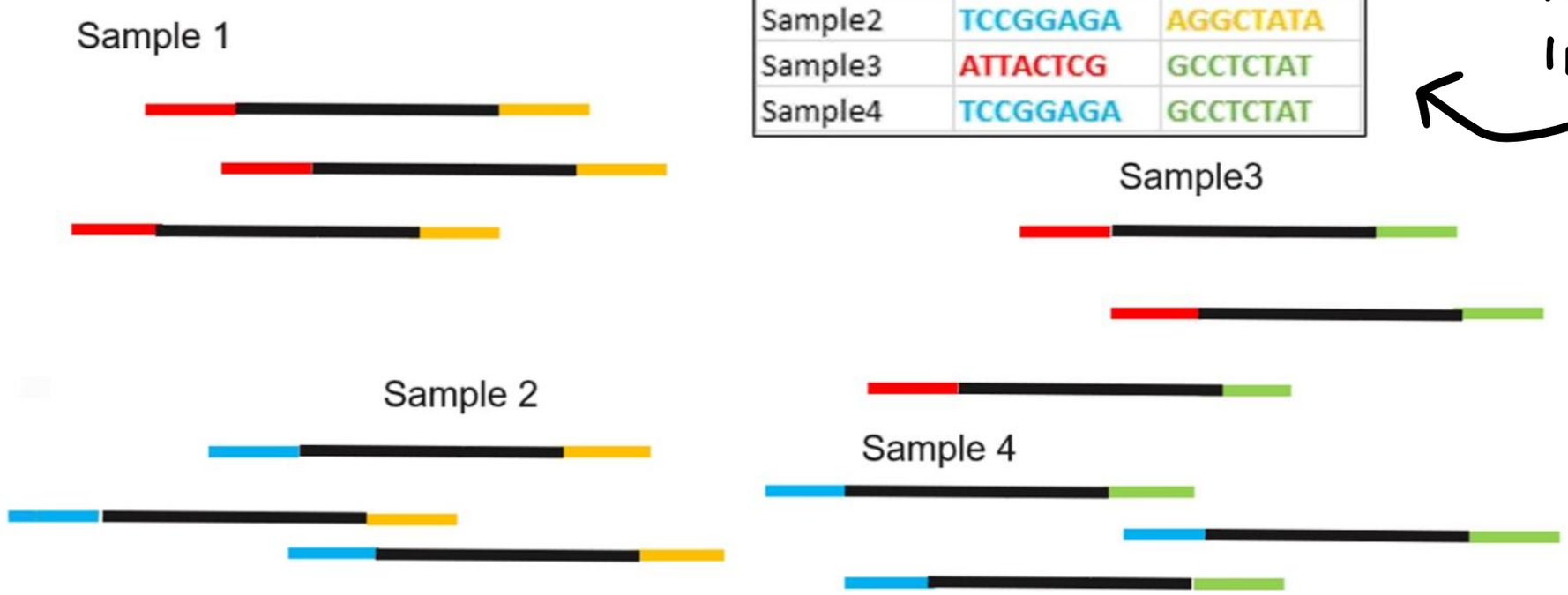
Demultiplexing

What is demultiplexing?

- When reads coming off the chip are distinguished by their specific identifying DNA tags (index sequence) and organized by sample.

[Data]		
Sample_ID	index	index2
Sample1	ATTACTCG	AGGCTATA
Sample2	TCCGGAGA	AGGCTATA
Sample3	ATTACTCG	GCCTCTAT
Sample4	TCCGGAGA	GCCTCTAT

These samples have a Dual Index



For Research Use Only. Not for use in diagnostic procedures.

illumina

Most sequencing facilities will demultiplex for you and give you individual files for each sample!

Downloading Raw Data

- What are important considerations for downloading data from sources like NCBI?

NCBI SRA

www.ncbi.nlm.nih.gov/sra

SRA | SRA | syngnathus | Search

Create alert Advanced Help

Access
Public (3,457)

Source
DNA (2,780)
RNA (660)

RNA

Type
genome (670)

paired

Library Layout
paired (2,011)
single (1,448)

Platform
ABI SOLiD (7)
BGISEQ (8)
Illumina (3,210)
LS454 (5)
PacBio SMRT (39)

Strategy
EpiGenomics (6)
Exome (31)
Genome (678)
other (2,744)

Data in Cloud
GS (3,457)
S3 (3,457)

File Type
bam (57)
cram (11)
fastq (3,332)
illumina native (3)

fastq

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Summary 20 per page Send to:

View results as an expanded interactive table using the RunSelector. Send results to Run selector

Search results

Items: 1 to 20 of 3459 << First < Prev Page 1 of 173 Next > Last >>

1. Genome skim of **Syngnathus** fuscus VIMS:15222
   1 ILLUMINA (Illumina NovaSeq X Plus) run: 11M spots, 3.3G bases, 1.1Gb downloads
   Accession: SRX31263371
2. Genome skim of **Syngnathus** louisianae VIMS:15082
   1 ILLUMINA (Illumina NovaSeq X Plus) run: 11.9M spots, 3.6G bases, 1.2Gb downloads
   Accession: SRX31263122
3. Genome skim of **Syngnathus** louisianae USNM:FISH:477784
   1 ILLUMINA (Illumina NovaSeq X Plus) run: 14.3M spots, 4.3G bases, 1.3Gb downloads
   Accession: SRX30970959
4. Genome skim of **Syngnathus** fuscus USNM:FISH:477789
   1 ILLUMINA (Illumina NovaSeq X Plus) run: 20.8M spots, 6.3G bases, 1.9Gb downloads
   Accession: SRX30970953
5. Genome skim of **Syngnathus** fuscus USNM:FISH:477792
   1 ILLUMINA (Illumina NovaSeq X Plus) run: 24M spots, 7.3G bases, 2.2Gb downloads
   Accession: SRX30970950
6. Genome skim of **Syngnathus** fuscus USNM:FISH:477880
   1 ILLUMINA (Illumina NovaSeq X Plus) run: 19.9M spots, 6G bases, 1.8Gb downloads
   Accession: SRX30970942
7. RNA-Seq of **Syngnathus** leptorhynchus - Adult Female Ovary
   1 ILLUMINA (Illumina NovaSeq 6000) run: 47.1M spots, 15G bases, 4.9Gb downloads
   Accession: SRX30264767

RNA-Seq of ASyngnathus leptorhynchus - Adult Male Testes

Filters: Manage Filters

Results by taxon

Top Organisms [Tree]
- Syngnathus scovelli *(1426)*
- Syngnathus typhle *(611)*
- fish metagenome *(537)*
- gut metagenome *(380)*
- Vibrio alginolyticus *(115)*
- All other taxa *(390)*

More...

Species

Search in related databases

Database	Access		all
	public	controlled	
BioSample	2,817		2,817
BioProject	53		53
dbGaP			
GEO Datasets	3		3

Find related data

Database: Select

Find items

Search details

```
"Syngnathus"[Organism] OR
syngnathus[All Fields]
```

Search See more...

NCBI SRA

www.ncbi.nlm.nih.gov/sra

how was the sample collected

what was the sequencing strategy

Full ▾ Send to: ▾

SRX30264767: **RNA-Seq of Syngnathus leptorhynchus - Adult Female Ovary**
1 ILLUMINA (Illumina NovaSeq 6000) run: 47.1M spots, 15G bases, 4.9Gb downloads

Design: Paired End RNA-Seq reads generated by polyA selection from various tissue for Genome Annotation

Submitted by: University of Idaho at Moscow

Study: The evo-devo of male pregnancy and its effects on the brood pouch microbiome in syngnathid fishes
PRJNA851781 • SRP383388 • All experiments • All runs
hide Abstract

The evolution of novel traits can change the way that organisms interact with their environments to survive, grow and reproduce. Deep knowledge of the underlying genes and developmental changes that underly most evolutionary innovations is sparse, as is the understanding of the ecological consequences for both the organisms in which novel traits emerged and the organisms with which they interact in communities. A particular gap in understanding is how the evolution of novel traits influences the biodiversity of their associated microbial communities. This project will help fill this gap in our knowledge by studying a remarkable innovation of male pregnancy in seahorses, pipefish, and seadragons. This project will include the creation of new genome sequences and detailed studies of the developmental genetic underpinnings of the embryo brooding structures that make male pregnancy possible. The consequences of pouch evolution on the complexity and function of the microbial communities in the pouch will also be studied, as well as how this unique host-associated microbiota can affect the fitness of embryos in the pouch.

Sample: RNA-Seq of Syngnathus leptorhynchus - Adult Female Ovary
SAMN50671166 • SRS26315687 • All experiments • All runs
Organism: Syngnathus leptorhynchus

Library:
Name: Slep_ovary
Instrument: Illumina NovaSeq 6000
Strategy: RNA-Seq
Source: TRANSCRIPTOMIC
Selection: PolyA
Layout: PAIRED

Runs: 1 run, 47.1M spots, 15G bases, 4.9Gb

Run	# of Spots	# of Bases	Size	Published
SRR35159596	47,080,517	15G	4.9Gb	2025-08-27

ID: 40401502

NCBI SRA

www.ncbi.nlm.nih.gov/sra

Full ▾ Send to: ▾

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sample info

Sample: RNA-Seq of Syngnathus leptorhynchus - Adult Female Ovary
SAMN50671166 • SRS26315687 • All experiments • All runs
Organism: Syngnathus leptorhynchus

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info to download reads

Runs: 1 run, 47.1M spots, 15G bases, 4.9Gb

Run	# of Spots	# of Bases	Size	Published
SRR35159596	47,080,517	15G	4.9Gb	2025-08-27

ID: 40401502

NCBI SRA

www.ncbi.nlm.nih.gov/sra

more info on the sample

more samples from the project

RNA-Seq of Syngnathus leptorhynchus - Adult Female Ovary

Identifiers BioSample: SAMN50671166; Sample name: Slep_ovary; SRA: SRS26315687

Organism Syngnathus leptorhynchus (bay pipefish)
cellular organisms; Eukaryota; Opisthokonta; Metazoa; Eumetazoa; Bilateria; Deuterostomia; Chordata; Craniata; Vertebrata; Gnathostomata; Teleostomi; Euteleostomi; Actinopterygii; Actinopteri; Neopterygii; Teleostei; Osteoglossocephalai; Clupeocephala; Euteleosteomorpha; Neoteleostei; Eurypterygia; Ctenosquamata; Acanthomorphata; Euacanthomorphacea; Percomorphaceae; Syngnathiaria; Syngnathiformes; Syngnathoidei; Syngnathidae; Syngnathinae; Syngnathus

Package Model organism or animal; version 1.0

Attributes

isolate	USA:Oregon
development stage	Adult
collection date	2018-08-16
geographic location	USA:Oregon
sex	female
tissue	ovary
isolation source	wild-caught

BioProject PRJNA851781 Doryrhamphus excisus
Retrieve all samples from this project

Submission University of Idaho at Moscow, Mark Currey; 2025-08-18

Accession: SAMN50671166 ID: 50671166
BioProject SRA

NCBI SRA

www.ncbi.nlm.nih.gov/sra

RNA-Seq of Syngnathus leptorhynchus - Adult Female Ovary (SRR35159596)

Metadata | Analysis | Reads | Data access | FASTA/FASTQ download

download these reads here or using SRA Toolkit

Run

Run	Spots	Bases	Size	GC Content	Data Status	Published
SRR35159596	47.1M	15.0G	4.9GB	49.1%	Public	2025-08-27

Quality graph (bigger)

This run has 2 reads per spot:

L=159, 100%	L=159, 100%

Legend

159 bp paired-end

Experiment

Experiment	Library Name	Platform	Strategy	Source	Selection	Layout	Action
SRX30264767	Slep_ovary	Illumina	RNA-Seq	TRANSCRIPTOMIC	PolyA	PAIRED	BLAST

poly A selection (reduces rRNA)

Design:
Paired End RNA-Seq reads generated by polyA selection from various tissue for Genome Annotation

Biosample

Biosample	Sample Description	Organism	Links
SAMN50671166 (SRS26315687)		Syngnathus leptorhynchus	The evo-devo of male pregnancy and its effects o

Bioproject

Bioproject	SRA Study	Title
PRJNA851781	SRP383388	The evo-devo of male pregnancy and its effects on the brood pouch microbiome in syngnathid fishes

Show abstract

NCBI SRA

www.ncbi.nlm.nih.gov/sra

For this workshop, I have already downloaded some data for us to work with!

159 bp paired-end

poly A selection (reduces rRNA)

RNA-Seq of Syngnathus leptorhynchus - Adult Female Ovary (SRR35159596)

Metadata | Analysis | Reads | Data access | FASTA/FASTQ download

Run

Run	Spots	Bases	Size	GC Content	Data Status	Published
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PRJNA851781	SRP383388	The evo-devo of male pregnancy and its effects on the brood pouch microbiome in syngnathid fishes

Show abstract

Using the Cluster

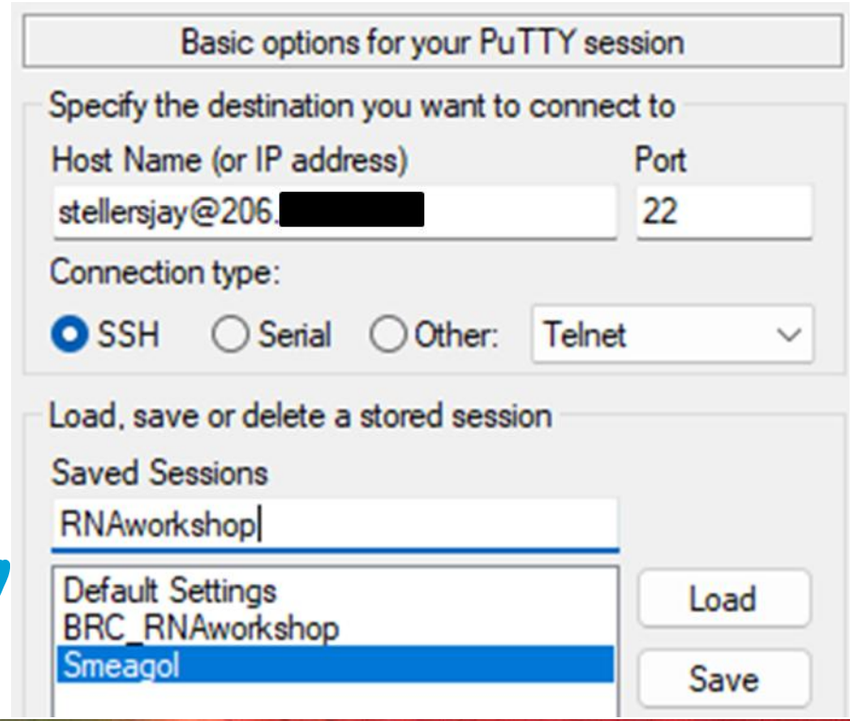
- Log into ARC Sockeye Accounts
- Learning to Navigate the Command Line with Bash
- Bash game!

Log into ARC Sockeye Accounts

- Reference handouts for account information
- Mac users can ssh from their terminal

```
guppybyte@GuppyByte:~$ ssh stellersjay@206.[redacted]
```

- Windows users can use PuTTY



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Learning Bash

- Default when using Linux (what is used for most supercomputers)
- Scripting language = short and simple set of instructions that tells the computer to automate processes

Our goal is to learn enough to run the programs we are interested in and maybe pick up some little tricks to make our lives easier!

Learning to Navigate the Command Line

- Let's start by practicing some of the commands on the cheat sheet!

```
> ls                    # list what is in the current directory
```

This represents typing in the terminal.

This is a command.

This is a comment. (It's your notes, not something you need to type into the computer).

Learning to Navigate the Command Line

- Commands are usually shorthand for their meaning.

```
> ls        # list
> cd        # change directory
> mkdir     # make directory
> cp        # copy
> rm        # remove
```

Bash game!

- Let’s continue our practicing by playing a game

```
> ls                          # look around the current directory
> cd workshop                 # navigate into the folder called workshop
> cd day1                     # navigate into the folder called day1
> ls                          # look around the current directory
> cp 01_bashgame ~/           # copy file into your home directory
> chmod u+x 01_bashgame       # change permissions of the file
> ./01_bashgame               # run the script so it can generate the game
```

When you finish give me a thumbs up!

Assessing Quality & Trimming

- How can I check the quality of my reads?
- What do I need to trim out of my data?

Assessing Quality & Trimming

How can I check the quality of my reads?

- FastQC

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Input: fastq file → **Output: results.html**

- per base sequence quality
- per sequence quality score
- overrepresented sequences
- adapter content

FastQC

Input: fastq file

→

Output: results.html

- per base sequence quality
- per sequence quality score
- overrepresented sequences
- adapter content

> fastqc --help # this will give you the manual

```
ubuntu@bioinformatics-[redacted]:~$ fastqc --help

            FastQC - A high throughput sequence QC analysis tool

SYNOPSIS

        fastqc seqfile1 seqfile2 .. seqfileN
```

FastQC

Input: fastq file → **Output: results.html**

- per base sequence quality
- per sequence quality score
- overrepresented sequences
- adapter content

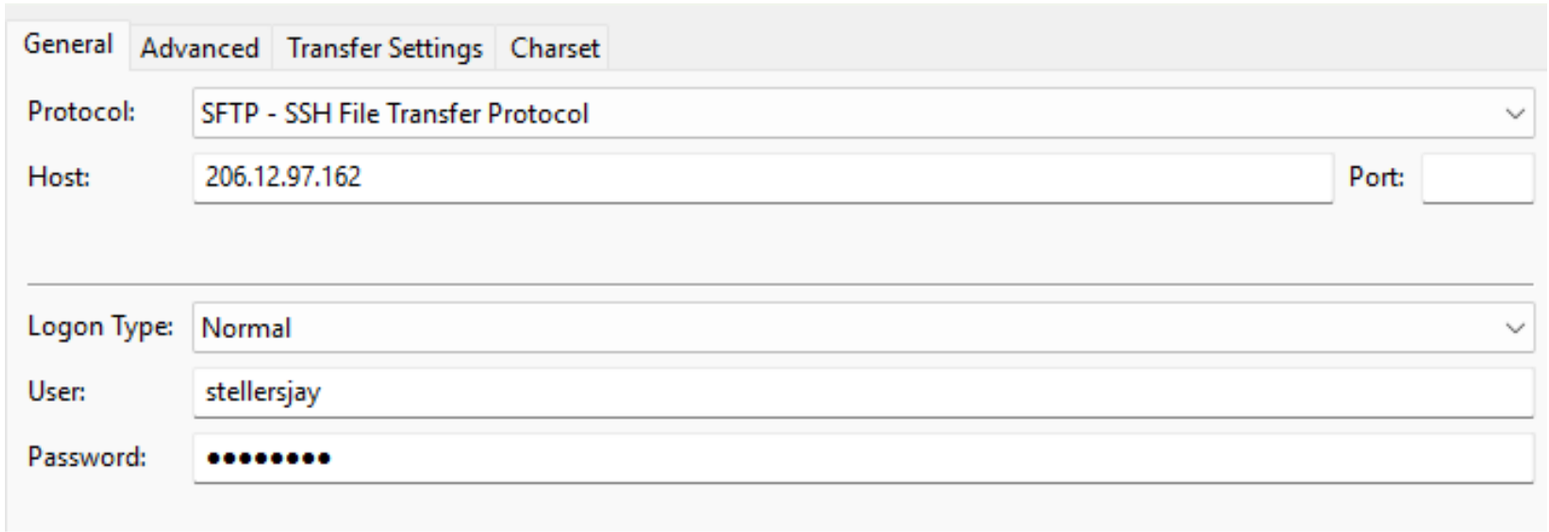
Let's run our first program!

1. Start by locating an example fastq file provided for you. It should be in your ***workshop*** directory.
2. Copy over the files into your home directory.
3. Before running FastQC, lets open a ***screen*** session.
4. Run FastQC on this file in your screen session.
5. For practice, try ***disconnecting*** from your screen session.
6. Check if your program is running using ***htop***.
7. Close htop by pressing 'q'.

Learning how to transfer files

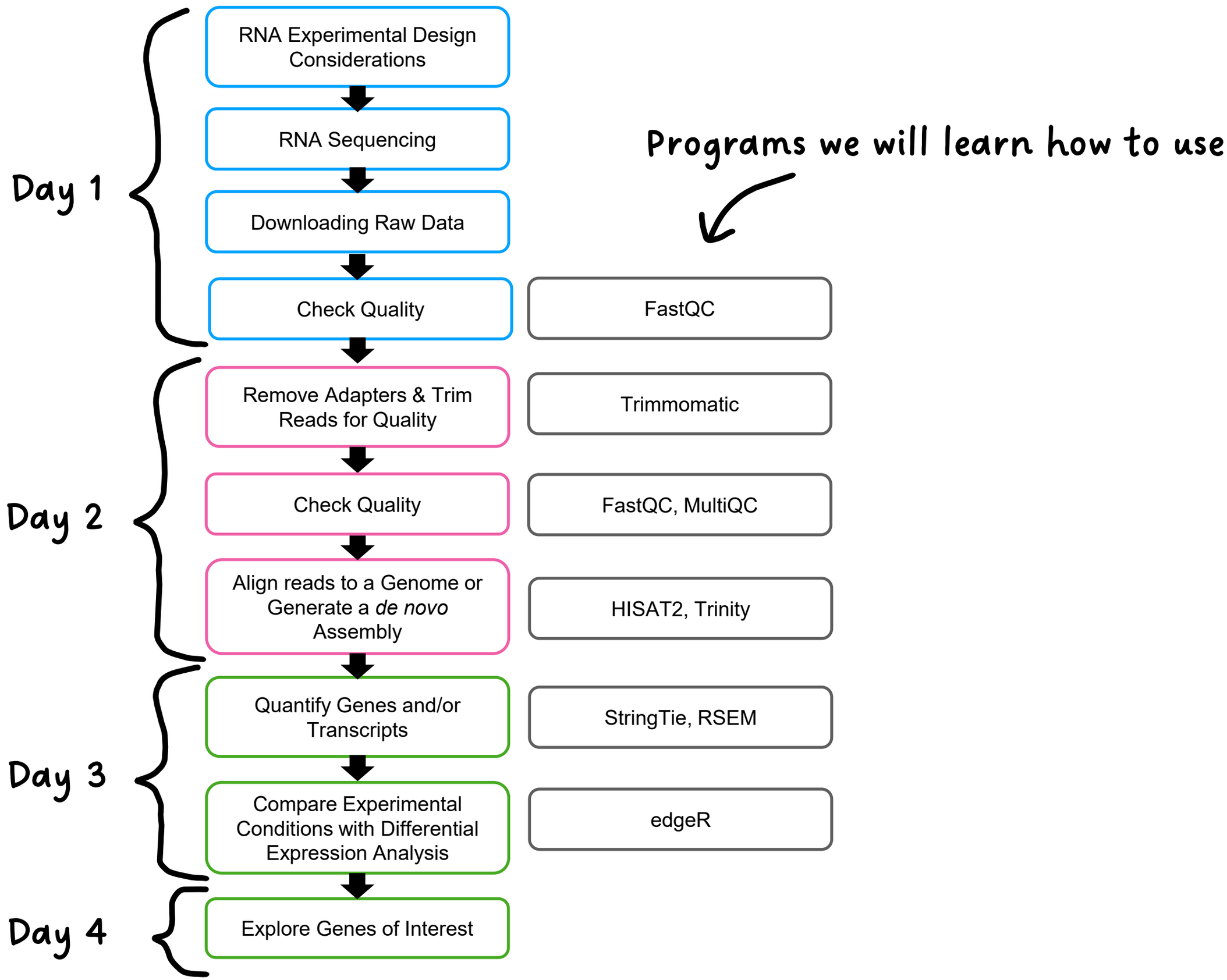
Let's download our results!

1. Make sure FastQC is done running.
2. Exit the screen session.
3. List the contents of the working directory.
4. Let's use FileZilla to transfer files over to our local computer.



5. We can open the html files from our local computer and view the results… next time!

Basic RNA-seq pipeline



Course Info

- Course Materials on Github: https://github.com/BernadetteBiology
- Photos, unless otherwise noted, are under © Bernadette J, some rights reserved (CC-BY-NC)